

VERILUMEN / COMPTY ? Complete Prompt Archive

****Generated:**** 2026-07-07

****Repositories:****

- Frontend

Part A ? STEP Prompt Archive (prompts.csv)

Total entries: ****54****

STEP 1 ? Project Setup

****Date:**** 2026-06-29

Prompt

Create Next.js app with TypeScript Tailwind ESLint App Router src directory. Install recharts framer-motion react-icons lucide-react react-query clsx tailwind-merge. Init shadcn and add button card table dropdown-menu input select slider avatar badge.

****Files:**** package.json; components.json; src/app/layout.tsx; src/app/globals.css; src/lib/utils.ts; src/components/ui/*.tsx

****Components:**** shadcn UI primitives (Button Card Table DropdownMenu Input Select Slider Avatar Badge)

STEP 2 ? Folder Structure

****Date:**** 2026-06-29

Prompt

Define src folder structure for dashboard components layout cards charts tables filters optimization results lib types hooks styles.

****Files:**** src/types/dashboard.ts; src/lib/dummyData.ts; src/styles/globals.css

****Components:**** Project scaffolding folders

STEP 3 ? Layout (Cursor Prompt 1)

****Date:**** 2026-06-29

Prompt

Premium enterprise dashboard layout: 280px sidebar 72px navbar 24px padding/gap CSS Grid glassmorphism purple accent dark theme responsive. Background #090B12 Cards #111827 Border #2D3748 Accent #7C3AED Rounded 20px.

****Files:**** src/components/layout/DashboardLayout.tsx; src/components/layout/Sidebar.tsx; src/components/layout/TopNavbar.tsx; src/styles/globals.css; src/app/globals.css

****Components:**** DashboardLayout Sidebar TopNavbar

STEP 4 ? Sidebar (Cursor Prompt 2)

****Date:**** 2026-06-29

Prompt

280px sidebar #0A1020 background. Header ATE Intelligence Enterprise Platform. Navigation: Dashboard Scan Chain Analysis MBIST LBIST Wafer Analysis Cost Intelligence Alerts Settings. Active menu purple gradient rounded-xl glow. Icons for every menu. Quick Filters: Date Range Fab Tester Product Reset Filters. Alerts badge 5. Sidebar fixed.

****Files:**** src/components/layout/Sidebar.tsx

****Components:**** Sidebar with navigation and Quick Filters

STEP 5 ? Top Navbar (Cursor Prompt 3)

****Date:**** 2026-06-29

Prompt

72px sticky navbar. Left page title Executive Dashboard. Center large search bar. Right Calendar Notifications Profile Export Report AI Optimize. User Alex Johnson Admin. Notification badge 12. Glass backdrop blur.

****Files:**** src/components/layout/TopNavbar.tsx

****Components:**** TopNavbar

STEP 6 ? Executive KPI Cards (Cursor Prompt 4)

****Date:**** 2026-06-29

Prompt

Six KPI cards in 6-column grid. Each card: icon title large value weekly trend sparkline hover animation. Metrics: Total Test Cost Cost per Wafer Cost per Die Test Time Yield ROI Improvement. Recharts AreaChart sparklines Framer Motion entrance. Glass card gradient border.

****Files:**** src/components/cards/ExecutiveCard.tsx; src/lib/dummyData.ts (executiveKPIs); src/types/dashboard.ts (ExecutiveKPI)

****Components:**** ExecutiveCard ExecutiveKPIGrid

STEP 7 ? Wafer Heatmap (Cursor Prompt 5)

****Date:**** 2026-06-29

Prompt

Wafer Cost Heatmap canvas 40x40 circular wafer grid. Pan zoom reset fullscreen. Overlay dropdown fail density yield cost. Color legend green yellow orange red. Tooltip on hover. Spatial AI analysis.

****Files:**** src/components/charts/WaferHeatmap.tsx; src/lib/dummyData.ts (generateWaferHeatData); src/types/dashboard.ts (HeatmapOverlay)

****Components:**** WaferHeatmap

STEP 8 ? Pattern Analysis Table (Cursor Prompt 6)

****Date:**** 2026-06-29

Prompt

Enterprise table columns: Pattern ID Test Time Cost Fail Rate Detect Power ROI Score Recommendation. Badges Keep Review Remove. Sticky header pagination search sorting hover row highlight.

****Files:**** src/components/tables/PatternTable.tsx; src/lib/dummyData.ts (patternAnalysisData); src/types/dashboard.ts (PatternRow Recommendation)

****Components:**** PatternTable

STEP 9 ? Cost Trend Chart (Cursor Prompt 7)

****Date:**** 2026-06-29

Prompt

Recharts line chart Total Cost and Cost per Wafer over 7 days Mon-Sun. Smooth animated lines dark theme legend. Glass card wrapper.

****Files:**** src/components/charts/CostTrendChart.tsx; src/lib/dummyData.ts (costTrendData); src/types/dashboard.ts (CostTrendPoint)

****Components:**** CostTrendChart

STEP 10 ? Optimization Engine (Cursor Prompt 8)

Date: 2026-06-29

Prompt

Three sliders Maximum Cost Yield Target Maximum Test Time with live values. Run AI Optimization purple button with sparkle icon animated loading state.

Files: src/components/optimization/OptimizationEngine.tsx; src/types/dashboard.ts (OptimizationParams)

Components: OptimizationEngine

STEP 11 ? Optimization Results (Cursor Prompt 9)

Date: 2026-06-29

Prompt

Result card after optimization: cost reduction time savings projected yield patterns reduced total savings. Green positive values. View Optimized Pattern Set button hover animation.

Files: src/components/results/OptimizationResult.tsx; src/lib/dummyData.ts (defaultOptimizationResults); src/types/dashboard.ts (OptimizationResults)

Components: OptimizationResult

STEP 12 ? Final Polish (Cursor Prompt 10)

Date: 2026-06-29

Prompt

Inter font glassmorphism gradient borders hover lift responsive desktop tablet mobile. Integrate all components dummy JSON production ready enterprise SaaS quality comparable to Synopsys Siemens NVIDIA Intel dashboards.

Files: src/app/dashboard/page.tsx; src/app/layout.tsx; src/app/globals.css; src/styles/globals.css; src/contexts/ThemeContext.tsx

Components: Full dashboard integration and polish

SETTINGS ? Settings Page

Date: 2026-06-29

Prompt

Theme Settings: appearance accent sidebar card font compact animations reset. Account Presets: profile role department dashboard language timezone notifications save. Persist to localStorage. Live theme preview.

Files: src/app/settings/page.tsx; src/types/theme.ts; src/contexts/ThemeContext.tsx; src/components/ui/switch.tsx; src/components/ui/radio-group.tsx; src/components/ui/label.tsx; src/components/settings/*.tsx

Components: Settings page ThemeProvider ThemeSettingsPanel AccountPresetsPanel

INTEGRATION ? Dashboard Page Integration

Date: 2026-06-29

Prompt

Wire all components into main dashboard page with shared dummy data and responsive grid layout. Redirect root / to /dashboard.

Files: src/app/dashboard/page.tsx; src/app/page.tsx

Components: DashboardPage

STEP 13 ? Scan Chain Analysis Dashboard

****Date:**** 2026-06-29

Prompt

Cursor AI Prompt ? Scan Chain Analysis Dashboard. Create premium enterprise Scan Chain Analysis dashboard for ATE Intelligence Enterprise Platform. Stack: Next.js 15 TypeScript TailwindCSS shadcn/ui Framer Motion Recharts Lucide. Theme: Background #090B12 Cards #111827 Border #2D3748 Accent #7C3AED Rounded 20px Glass Effect smooth animations responsive. LEFT SIDEBAR: ATE Intelligence Enterprise Platform nav Dashboard Scan Chain Analysis (Active) MBIST LBIST Wafer Cost Intelligence Alerts Settings purple gradient active glow fixed. TOP NAV: Scan Chain Analysis title search placeholder Search scan chains patterns chips flops Calendar Notifications Profile Export Report AI Diagnose sticky. SECONDARY NAV tabs below title: Overview Pattern Analysis Failure Analysis Scan Diagnosis Overview active purple underline bold smooth tab switch content. OVERVIEW TAB: 6 KPI cards Total Scan Chains Failing Scan Chains Failing Flops Scan Coverage Average Test Time Pattern Count icon title value trend sparkline hover. Row 2: Scan Chain Health Summary donut Healthy Warning Failing Unknown center Total Chains; Top Failing Chips horizontal bar top 10; Scan Chain Heatmap grid green yellow orange red legend. Row 3: Recent Failing Scan Chains table Chain ID Pattern ID Chip Fail Cycle Fail Type Suspected Root Cause Diagnosis Status Action search pagination sorting; Failure Distribution pie Stuck At Transition Bridging Timing Unknown; AI Diagnosis Summary card root cause critical chains debug area repair success confidence Run AI Diagnosis button. PATTERN ANALYSIS TAB: KPIs Total Patterns Active Patterns Pattern Coverage Pattern Efficiency Pattern Runtime Compressed Patterns; charts Pattern Execution Trend Pattern Cost Pattern Coverage Pattern Density; tables Pattern Summary Pattern Recommendations. FAILURE ANALYSIS TAB: KPIs Total Failures Critical Warning Recovered Root Causes Repair Rate; charts Failure Trend Failure Type Distribution Failing Regions Failure Density; tables Failure Records Root Cause Analysis AI Recommendation. SCAN DIAGNOSIS TAB: KPIs Diagnosed Unknown Repair Suggestions Diagnosis Accuracy Coverage Confidence; charts Diagnosis Timeline Chain Connectivity Graph Failure Propagation AI Confidence; tables Diagnosis Report Suspected Scan Cells Recommended Debug Points Repair Priority. COMMON: glass cards border hover lift shadow rounded 20px 24px spacing grid 2xl desktop 2 col tablet 1 col mobile. Animations fade slide scale hover glow sparkline. Typography Inter bold white headers gray subtitles large colored values. Reusable components dummy JSON separate components clean folder structure responsive production ready enterprise SaaS Synopsys Siemens NVIDIA Intel quality.

****Files:**** src/app/scan-chain/page.tsx; src/types/scanChain.ts; src/lib/scanChainData.ts; src/components/scan-chain/KPICard.tsx; src/components/scan-chain/ScanChainTabs.tsx; src/components/scan-chain/ChartCard.tsx; src/components/scan-chain/DataTable.tsx; src/components/scan-chain/ScanChainHeatmap.tsx; src/components/scan-chain/AIDiagnosisCard.tsx; src/components/scan-chain/ConnectivityGraph.tsx; src/components/scan-chain/charts/PieCharts.tsx; src/components/scan-chain/charts/BarCharts.tsx; src/components/scan-chain/charts/LineCharts.tsx; src/components/scan-chain/tabs/OverviewTab.tsx; src/components/scan-chain/tabs/PatternAnalysisTab.tsx; src/components/scan-chain/tabs/FailureAnalysisTab.tsx; src/components/scan-chain/tabs/ScanDiagnosisTab.tsx; src/components/layout/Sidebar.tsx; src/components/layout/TopNavbar.tsx; src/components/layout/DashboardLayout.tsx

****Components:**** KPICard KPISummary ScanChainTabs ChartCard DataTable ScanChainHeatmap AIDiagnosisCard ConnectivityGraph DonutChart DistributionPie HorizontalBarChart VerticalBarChart TrendLineChart TrendAreaChart OverviewTab PatternAnalysisTab FailureAnalysisTab ScanDiagnosisTab

STEP 14 ? Record All Prompts

****Date:**** 2026-06-29

Prompt

Record the all prompt in csv and readme file.

****Files:**** prompts.csv; README.md

****Components:**** Prompt documentation archive

STEP 15 ? Automatic Prompt Recording

****Date:**** 2026-06-29

Prompt

make the automatic record the prompt in csv and readme file.

****Files:**** .cursor/hooks.json; .cursor/hooks/prompt-recorder.mjs; .cursor/hooks/record-prompt-start.mjs; .cursor/hooks/record-prompt-stop.mjs; .cursor/hooks/track-file-edit.mjs; scripts/record-prompt.mjs; package.json; .gitignore
****Components:**** prompt-recorder record-prompt-start record-prompt-stop track-file-edit

STEP 16 ? Recommendation Analysis Sidebar

****Date:**** 2026-06-29

Prompt

Add Recommendation Analysis to sidebar between Cost Intelligence and Alerts. Sparkles icon route /dashboard/recommendation-analysis. Purple gradient active state hover scale accessibility. Do not redesign sidebar.

****Files:**** src/components/layout/Sidebar.tsx; src/app/dashboard/recommendation-analysis/page.tsx
****Components:**** Sidebar RecommendationAnalysisPage

STEP 17 ? MBIST Analysis Dashboard

****Date:**** 2026-06-29

Prompt

Create premium enterprise MBIST Analysis dashboard. Stack Next.js TypeScript Tailwind shadcn Framer Motion Recharts Lucide. Theme dark enterprise glassmorphism. Sidebar MBIST Analysis active. Header MBIST Analysis subtitle Memory Built-In Self-Test Analytics. Tabs Overview Memory Health Failure Analysis Diagnosis AI Recommendation. Overview 6 KPIs memory health donut failure by bank heatmap failures table failure pie AI diagnosis. Memory Health KPIs utilization temperature access density charts. Failure Analysis KPIs trend type bank heatmap records table. Diagnosis KPIs timeline correlation connectivity root cause graphs diagnosis fail address repair tables. AI Recommendation risk cards recommendations table. Route /mbist.

****Files:**** src/app/mbist/page.tsx; src/types/mbist.ts; src/lib/mbistData.ts; src/components/mbist/KPICard.tsx; src/components/mbist/MBISTTabs.tsx; src/components/mbist/MBISTHeatmap.tsx; src/components/mbist/MBISTAIAnalysisCard.tsx; src/components/mbist/MemoryGraphs.tsx; src/components/mbist/tabs/OverviewTab.tsx; src/components/mbist/tabs/MemoryHealthTab.tsx; src/components/mbist/tabs/FailureAnalysisTab.tsx; src/components/mbist/tabs/DiagnosisTab.tsx; src/components/mbist/tabs/AIRecommendationTab.tsx; src/components/layout/Sidebar.tsx; src/components/layout/TopNavbar.tsx; src/components/layout/DashboardLayout.tsx
****Components:**** MBISTTabs KPIGrid MBISTHeatmap MBISTAIAnalysisCard MemoryConnectivityGraph RootCauseGraph OverviewTab MemoryHealthTab FailureAnalysisTab DiagnosisTab AIRecommendationTab

STEP 18 ? LBIST Analysis Dashboard

****Date:**** 2026-06-29

Prompt

Create premium enterprise LBIST Analysis dashboard. Subtitle Logic Built-In Self-Test Analytics Coverage and Diagnosis. Tabs Overview Coverage Analysis Failure Analysis Diagnosis AI Recommendation. Overview 6 KPIs coverage donut failure by module heatmap failures table failure pie AI diagnosis. Coverage Analysis KPIs trend block efficiency detection heatmap. Failure Analysis KPIs trend distribution density records tables. Diagnosis timeline correlation connectivity coverage correlation tables AI diagnosis. AI Recommendation 7 risk cards recommendations table. Route /lbist. Same design as Executive Scan Chain MBIST dashboards.

****Files:**** src/app/lbist/page.tsx; src/types/lbist.ts; src/lib/lbistData.ts; src/components/lbist/KPICard.tsx; src/components/lbist/LBISTTabs.tsx; src/components/lbist/LBISTHeatmap.tsx; src/components/lbist/LBISTAIAnalysisCard.tsx; src/components/lbist/LogicGraphs.tsx; src/components/lbist/tabs/OverviewTab.tsx; src/components/lbist/tabs/CoverageAnalysisTab.tsx; src/components/lbist/tabs/FailureAnalysisTab.tsx; src/components/lbist/tabs/DiagnosisTab.tsx; src/components/lbist/tabs/AIRecommendationTab.tsx; src/components/layout/Sidebar.tsx
****Components:**** LBISTTabs KPIGrid LBISTHeatmap LBISTAIAnalysisCard LogicConnectivityGraph OverviewTab CoverageAnalysisTab FailureAnalysisTab DiagnosisTab AIRecommendationTab

STEP 19 ? Recommendation Analysis Dashboard

Date: 2026-06-29

Prompt

Create premium enterprise Recommendation Analysis dashboard centralized AI recommendation module consolidating Scan Chain MBIST LBIST and Wafer Analysis. Route /recommendation-analysis. Header subtitle AI-powered unified recommendations. Primary action Generate AI Recommendations. Tabs Overview Scan Chain MBIST LBIST Wafer Analysis. Overview 6 KPIs source donut priority bar trend line unified table AI executive summary recommendation engine workflow bottom AI summary. Module tabs domain KPIs charts tables. Priority badges Critical High Medium Low. Dark enterprise glassmorphism theme.

Files: src/app/recommendation-analysis/page.tsx; src/types/recommendation.ts;
src/lib/recommendationData.ts; src/components/recommendation/KPICard.tsx;
src/components/recommendation/RecommendationTabs.tsx; src/components/recommendation/Badges.tsx;
src/components/recommendation/AIExecutiveSummaryCard.tsx; src/components/recommendation/EnginePanels.tsx;
src/components/recommendation/WaferRecHeatmap.tsx; src/components/recommendation/tabs/OverviewTab.tsx;
src/components/recommendation/tabs/ScanChainTab.tsx; src/components/recommendation/tabs/MbistTab.tsx;
src/components/recommendation/tabs/LbistTab.tsx; src/components/recommendation/tabs/WaferTab.tsx;
src/components/layout/Sidebar.tsx; src/app/dashboard/recommendation-analysis/page.tsx
Components: RecommendationTabs KPIGrid PriorityBadge ModuleBadge StatusBadge AIExecutiveSummaryCard
RecommendationEnginePanel WorkflowPanel BottomAISummaryPanel OverviewTab ScanChainTab MbistTab LbistTab
WaferTab WaferRecHeatmap

STEP 20 ? Cost Intelligence Dashboard

Date: 2026-06-29

Prompt

Create premium enterprise Cost Intelligence dashboard. Route /cost-intelligence. Subtitle analyze and optimize semiconductor test costs across Scan Chain MBIST LBIST and Wafer Analysis. Primary action Generate Cost Optimization. Tabs Overview Scan Chain Cost MBIST Cost LBIST Cost Wafer Cost AI Cost Optimization. Overview 6 KPIs cost contribution donut cost breakdown bar monthly trend stacked distribution product cost table AI cost summary enterprise summary. Module tabs KPIs charts tables. AI optimization recommendation table engine categories enterprise cost summary.

Files: src/app/cost-intelligence/page.tsx; src/types/costIntelligence.ts; src/lib/costIntelligenceData.ts;
src/components/cost-intelligence/KPICard.tsx; src/components/cost-intelligence/CostTabs.tsx;
src/components/cost-intelligence/CostCharts.tsx; src/components/cost-intelligence/CostPanels.tsx;
src/components/cost-intelligence/WaferCostHeatmap.tsx; src/components/cost-intelligence/tabs/OverviewTab.tsx;
src/components/cost-intelligence/tabs/ScanChainCostTab.tsx; src/components/cost-intelligence/tabs/MbistCostTab.tsx;
src/components/cost-intelligence/tabs/LbistCostTab.tsx;
src/components/cost-intelligence/tabs/WaferCostTab.tsx; src/components/cost-intelligence/tabs/AICostOptimizationTab.tsx;
src/components/layout/Sidebar.tsx
Components: CostTabs KPIGrid ModuleHorizontalBarChart StackedCostBarChart AICostSummaryCard
CostOptimizationEnginePanel EnterpriseCostSummaryPanel OverviewTab ScanChainCostTab MbistCostTab LbistCostTab
WaferCostTab AICostOptimizationTab WaferCostHeatmap

STEP 21 ? Alerts Dashboard

Date: 2026-06-29

Prompt

Create premium enterprise Alerts dashboard real-time monitoring and notification center consolidating Scan Chain MBIST LBIST Wafer Cost Intelligence and AI Recommendation alerts. Route /alerts. Subtitle monitor and manage real-time alerts. Primary action Mark All as Read. Tabs Overview Scan Chain Alerts MBIST Alerts LBIST Alerts Wafer Alerts Cost Alerts AI Recommendation Alerts. Overview 6 KPIs alert distribution donut severity bar trend line recent alerts table critical alert summary workflow executive summary. Module tabs KPIs charts tables. Severity badges Critical High Medium Low. Alert workflow animated steps.

Files: src/app/alerts/page.tsx; src/types/alerts.ts; src/lib/alertsData.ts;

src/components/alerts/KPICard.tsx; src/components/alerts/AlertTabs.tsx; src/components/alerts/Badges.tsx;
src/components/alerts/AlertPanels.tsx; src/components/alerts/WaferAlertHeatmap.tsx;
src/components/alerts/tabs/OverviewTab.tsx; src/components/alerts/tabs/ScanChainAlertsTab.tsx;
src/components/alerts/tabs/MbistAlertsTab.tsx; src/components/alerts/tabs/LbistAlertsTab.tsx;
src/components/alerts/tabs/WaferAlertsTab.tsx; src/components/alerts/tabs/CostAlertsTab.tsx;
src/components/alerts/tabs/AIRecommendationAlertsTab.tsx; src/components/layout/Sidebar.tsx
****Components:**** AlertTabs KPIGrid SeverityBadge AlertStatusBadge ModuleBadge CriticalAlertSummaryCard
AlertWorkflowPanel ExecutiveAlertSummaryPanel OverviewTab ScanChainAlertsTab MbistAlertsTab LbistAlertsTab
WaferAlertsTab CostAlertsTab AIRecommendationAlertsTab WaferAlertHeatmap

STEP 22 ? Upload Test Data

****Date:**** 2026-06-29

Prompt

Add enterprise Upload Test Data feature top navbar UploadCloud button purple gradient modal drag-drop react-dropzone dataset category metadata progress upload history. Supported STDF STIL WGL CSV XLSX JSON ZIP XML max 10GB. Frontend UI UX only no backend simulated progress toast client-side history.

****Files:**** src/components/upload/UploadDataModal.tsx; src/components/upload/UploadDropzone.tsx;
src/components/upload/UploadProgressPanel.tsx; src/components/upload/UploadHistoryTable.tsx;
src/components/upload/UploadToastStack.tsx; src/contexts/UploadContext.tsx; src/types/upload.ts;
src/lib/uploadData.ts; src/components/ui/dialog.tsx; src/components/ui/progress.tsx;
src/components/layout/TopNavbar.tsx; src/app/layout.tsx
****Components:**** UploadDataModal UploadDropzone UploadProgressPanel DataUploadHistoryTable UploadProvider
UploadToastStack

STEP 23 ? Upload Log File

****Date:**** 2026-06-29

Prompt

Add enterprise Upload Log File button beside Upload Data FileText dark glass purple border modal ATE log upload log source tester metadata validation progress 5-step pipeline AI log summary upload history actions. Supported STDF STIL WGL LOG TXT CSV JSON XML ZIP GZ max 5GB. Frontend UI UX only no backend toast notifications.

****Files:**** src/components/upload/UploadLogFileModal.tsx; src/components/upload/UploadDataModal.tsx;
src/components/upload/UploadDropzone.tsx; src/components/upload/UploadProgressPanel.tsx;
src/components/upload/UploadHistoryTable.tsx; src/contexts/UploadContext.tsx; src/types/upload.ts;
src/lib/uploadData.ts; src/components/layout/TopNavbar.tsx
****Components:**** UploadLogFileModal LogUploadHistoryTable ValidationResult PipelineStep AILogSummary

STEP 24 ? Complete Remaining Platform Functionality

****Date:**** 2026-06-29

Prompt

Complete remaining frontend functionality for ATE Intelligence Enterprise Platform. Zustand stores filters user uploads notifications settings. Route migration to /dashboard/* with redirects and wafer-analysis page. Global search calendar notifications export responsive nav sidebar. Filter engine primary actions AI diagnosis apply recommendation. Upload persistence profile loading empty error states accessibility. Do not redesign page shell.

****Files:**** src/stores/*.ts; src/app/dashboard/wafer-analysis/page.tsx; src/components/filters/*;
src/components/search/*; src/components/notifications/*; src/components/export/*; src/components/ai/*;
src/components/states/*
****Components:**** FilterStore UserStore UploadStore NotificationStore SettingsStore GlobalSearch ExportPanel
AIDiagnosisPanel ApplyRecommendationButton

STEP 25 ? Pattern Analysis KPI Dashboard

****Date:**** 2026-06-29

Prompt

Scan Chain Pattern Analysis tab enterprise KPI dashboard. 11 KPIs PA-001 to PA-011: Total Patterns Active Patterns Pattern Coverage Pattern Efficiency Pattern Runtime Compressed Patterns Pattern Import Rate Pattern Redundancy Pattern Similarity Pattern Cluster Count Pattern Optimization Score. Charts: Pattern Import Trend Pattern Coverage Trend Pattern Cluster Distribution donut Pattern Similarity Scatter. Table: Pattern ID Category Coverage Efficiency Runtime Redundancy Similarity Cluster Recommendation Priority Status. Dark enterprise glass theme Recharts Framer Motion.

****Files:**** src/components/scan-chain/tabs/PatternAnalysisTab.tsx; src/components/scan-chain/pattern/PatternKPIGrid.tsx; src/components/scan-chain/pattern/PatternImportTrendChart.tsx; src/components/scan-chain/pattern/PatternCoverageTrendChart.tsx; src/components/scan-chain/pattern/PatternClusterPieChart.tsx; src/components/scan-chain/pattern/PatternSimilarityScatter.tsx; src/components/scan-chain/pattern/PatternAnalysisTable.tsx; src/types/scanChain.ts; src/lib/scanChainData.ts; src/styles/globals.css (.pattern-kpi-grid)

****Components:**** PatternKPIGrid PatternImportTrendChart PatternCoverageTrendChart PatternClusterPieChart PatternSimilarityScatter PatternAnalysisTable

STEP 26 ? Pattern Analysis Tab Refinement

****Date:**** 2026-06-29

Prompt

Remove from Scan Chain Pattern Analysis tab: header action box Upload Export Generate AI buttons AI Recommendation Summary section Redundancy Heatmap Similarity Matrix. Keep 11 KPIs import/coverage trend charts cluster pie scatter chart pattern table.

****Files:**** src/components/scan-chain/tabs/PatternAnalysisTab.tsx

****Components:**** PatternAnalysisTab (simplified layout)

STEP 27 ? Failure Analysis KPI Dashboard

****Date:**** 2026-06-29

Prompt

Scan Chain Failure Analysis tab enterprise KPI dashboard. 12 KPIs FA-FR-001 to FA-FR-012: Total Failures Critical Failures Warning Failures Recovered Failures Failure Rate Root Cause Count Repair Rate Mean Time to Repair Failure Density Failure Trend Score Failure Impact Score Failure Recovery Score. Charts: Failure Trend Failure Type Distribution Failure by Lot Failure Density. Table: Failure ID Pattern ID Chip Fail Type Root Cause Severity Status Repair Action Priority. Dark enterprise glass theme.

****Files:**** src/components/scan-chain/tabs/FailureAnalysisTab.tsx; src/components/scan-chain/failure/FailureKPIGrid.tsx; src/components/scan-chain/failure/FailureTrendChart.tsx; src/components/scan-chain/failure/FailureTypeDistributionChart.tsx; src/components/scan-chain/failure/FailureByLotChart.tsx; src/components/scan-chain/failure/FailureAnalysisTable.tsx; src/types/scanChain.ts; src/lib/scanChainData.ts

****Components:**** FailureKPIGrid FailureTrendChart FailureTypeDistributionChart FailureByLotChart FailureAnalysisTable

STEP 28 ? Failure Analysis Tab Refinement

****Date:**** 2026-06-29

Prompt

Remove from Scan Chain Failure Analysis tab: header action box Upload Export Generate AI buttons AI Recommendation Summary Wafer Die heatmaps Correlation Matrix Root Cause Analysis section. Keep 12 KPIs failure trend charts distribution by-lot charts failure table.

****Files:**** src/components/scan-chain/tabs/FailureAnalysisTab.tsx

****Components:**** FailureAnalysisTab (simplified layout)

STEP 29 ? Recommendation Analysis AI Agent Center

****Date:**** 2026-06-29

Prompt

Redesign Recommendation Analysis page as single AI Recommendation Center without module tabs Overview Scan Chain MBIST LBIST Wafer. Three agent sections on one page: Pattern Recommendation Agent Scan Debug Recommendation Agent Test Optimization Recommendation Agent. Each section KPIs charts tables AI summary workflow. Do not redesign sidebar or top nav.

****Files:**** src/app/dashboard/recommendation-analysis/page.tsx;

src/components/recommendation/RecommendationCenterContent.tsx;

src/components/recommendation/CenterKPIGrid.tsx; src/components/recommendation/AgentSummaryCard.tsx;

src/components/recommendation/AgentWorkflowDiagram.tsx; src/types/recommendation.ts;

src/lib/recommendationData.ts

****Components:**** RecommendationCenterContent CenterKPIGrid AgentSummaryCard AgentWorkflowDiagram

STEP 30 ? Recommendation Analysis AI Agent Tabs

****Date:**** 2026-06-29

Prompt

Redesign Recommendation Analysis with 3 AI Agent tabs persisted in uiStore: Pattern Recommendation Agent Scan Debug Recommendation Agent Test Optimization Recommendation Agent. Animated tab switching Framer Motion. Remove old 5 module tabs. Route /dashboard/recommendation-analysis.

****Files:**** src/components/recommendation/RecommendationCenterContent.tsx;

src/components/recommendation/AgentTabs.tsx; src/components/recommendation/tabs/PatternAgentTab.tsx;

src/components/recommendation/tabs/ScanDebugAgentTab.tsx;

src/components/recommendation/tabs/TestOptAgentTab.tsx; src/stores/uiStore.ts; src/types/recommendation.ts

****Components:**** AgentTabs PatternAgentTab ScanDebugAgentTab TestOptAgentTab RecommendationCenterContent

STEP 31 ? Scan Debug Recommendation Agent Tab

****Date:**** 2026-06-29

Prompt

Scan Debug Recommendation Agent tab full enterprise dashboard. 15 KPIs in 5 sections ai-rec-kpi-grid 5 cols desktop: Debug Queue Open Issues Critical Debug Items Root Cause Identified Repair Success Rate Diagnosis Coverage Debug Cycle Time Pattern Failures Scan Chain Failures Suspected Cells Debug Priority Score AI Confidence Debug Cost Impact Expected Yield Gain. Charts: root cause donut priority bar 30-day trend. Table: Recommendation ID Category Priority Expected Impact Action Status. AI Debug Executive Summary 9 cards. Debug workflow diagram action bar.

****Files:**** src/components/recommendation/tabs/ScanDebugAgentTab.tsx;

src/components/recommendation/SectionedKPIGrid.tsx; src/components/recommendation/AgentTabHeader.tsx;

src/components/recommendation/AgentActionBar.tsx; src/lib/recommendationData.ts (scanDebugAgent*);

src/styles/globals.css (.ai-rec-kpi-grid)

****Components:**** ScanDebugAgentTab SectionedKPIGrid AgentTabHeader AgentActionBar

STEP 32 ? Test Optimization Recommendation Agent Tab

****Date:**** 2026-06-29

Prompt

Test Optimization Recommendation Agent tab full enterprise dashboard. 19 KPIs in 7 sections test-opt-kpi-grid 3 cols desktop: Total Recommendations Active Optimizations Cost Savings Potential Time Savings Potential Yield

Improvement Potential Test Time Reduction Pattern Count Reduction Site Utilization Score Tester Efficiency
Score ROI Score Optimization Confidence Priority Score Assigned Engineers Pending Actions Completed Actions
Expected Cost Impact Expected Yield Gain Optimization Score. Charts: priority donut priority bar 30-day trend
yield line cost area site utilization heatmap 16 sites. Table: Recommendation ID Category Priority Confidence
Assigned Engineer Expected Impact Action Status. Executive AI Summary 8 cards. Optimization workflow action
bar.

****Files:**** src/components/recommendation/tabs/TestOptAgentTab.tsx;
src/components/recommendation/SiteUtilizationHeatmap.tsx; src/lib/recommendationData.ts (testOptAgent*);
src/styles/globals.css (.test-opt-kpi-grid)
****Components:**** TestOptAgentTab SiteUtilizationHeatmap SectionedKPIGrid AgentTabHeader AgentActionBar

STEP 33 ? Record All Prompts (Session Update)

****Date:**** 2026-06-29

Prompt

have upload all prompt to csv and readme file

****Files:**** prompts.csv; README.md

****Components:**** Prompt documentation archive (STEP 24-33)

STEP 34 ? Wafer Analysis Module (All Tabs)

****Date:**** 2026-06-29

Prompt

Create premium enterprise Wafer Analysis module for ATE Intelligence Enterprise Platform. Route
/dashboard/wafer-analysis redirect /wafer-analysis. Stack Next.js TypeScript Tailwind shadcn Recharts Framer
Motion Zustand HTML5 Canvas. Theme dark enterprise #090B12 #111827 #2D3748 #7C3AED glass 20px. Keep sidebar
top nav search upload export filters. Header Wafer Analysis subtitle AI-powered wafer defect classification
yield analysis hotspot detection spatial intelligence. Primary action Generate Yield Analysis. 10 tabs
TabPanelHost keep-alive: Overview Centre Donut Edge-Ring Scratch Near-Full Normal Edge-Loc Local Random.
Overview: Input Die Statistics 5 KPIs Defect Classification 9 clickable KPIs Yield Analysis positive negative
donut Gallery 9 cards Upload Workflow 8 steps Bottom Summary 7 KPIs. Defect tabs shared DefectClassTab: header
8 KPIs Upload History list Overlay Analytics canvas Fail Density canvas Info Panel workflow. Data waferData.ts
wafer.ts buildWaferImages getDefectBundle defectClassMeta galleryCards. Components WaferTabs
WaferNavigationContext KPICard DefectClassKPIGrid WaferGalleryGrid BottomSummaryBar UploadHistoryPanel
WaferAnalysisViews WaferInfoPanel OverviewTab DefectClassTab. Match Scan Chain MBIST enterprise quality.

****Files:**** src/app/dashboard/wafer-analysis/page.tsx; src/app/wafer-analysis/page.tsx; src/types/wafer.ts;
src/lib/waferData.ts; src/components/wafer/WaferTabs.tsx; src/components/wafer/WaferNavigationContext.tsx;
src/components/wafer/KPICard.tsx; src/components/wafer/DefectClassKPIGrid.tsx;
src/components/wafer/WaferGalleryGrid.tsx; src/components/wafer/BottomSummaryBar.tsx;
src/components/wafer/UploadHistoryPanel.tsx; src/components/wafer/WaferAnalysisViews.tsx;
src/components/wafer/WaferInfoPanel.tsx; src/components/wafer/tabs/OverviewTab.tsx;
src/components/wafer/tabs/DefectClassTab.tsx

****Components:**** WaferTabs WaferNavigationProvider TabPanelHost OverviewTab DefectClassTab KPIGrid
DefectClassKPIGrid WaferGalleryGrid BottomSummaryBar UploadHistoryPanel WaferAnalysisViews WaferInfoPanel

STEP 35 ? Wafer Analysis Images in Data

****Date:**** 2026-06-29

Prompt

Update wafer data to include wafer images in waferData.ts not separate asset system. WaferImages interface
wafer overlay density SVG data URIs via buildWaferImages buildWaferImageUri. Attach images to defectClassMeta
galleryCards topDefectWafers allUploads analysisRows. Components use img src from data. Remove separate
WaferDieMap canvas generator.

****Files:**** src/types/wafer.ts; src/lib/waferData.ts; src/components/wafer/WaferGalleryGrid.tsx;

src/components/wafer/UploadHistoryPanel.tsx; src/components/wafer/WaferAnalysisViews.tsx

Components: WaferImages buildWaferImages buildWaferImageUri

STEP 36 ? Wafer Analysis UI Refinement

Date: 2026-06-29

Prompt

Overview remove: Yield Trend Yield Distribution Defect Class Breakdown Recent Wafer Yield Top Defect Wafers table. Keep Input Die Statistics Defect Classification positive negative donut Gallery Workflow Summary. Defect tabs remove Class Probability Distribution AI Insights separate Recent History. Upload History single selectable list allUploads 10 items drives canvas maps. Overlay Analytics canvas teal good gold fail cyan cluster boxes. Fail Density canvas smooth heatmap blue cyan yellow purple crosshair X Y tooltip. Info panel updates from selected upload.

Files: src/components/wafer/tabs/OverviewTab.tsx; src/components/wafer/tabs/DefectClassTab.tsx; src/components/wafer/UploadHistoryPanel.tsx; src/components/wafer/WaferAnalysisViews.tsx; src/lib/waferData.ts; src/types/wafer.ts

Components: UploadHistoryPanel WaferAnalysisViews OverlayAnalyticsCanvas FailDensityCanvas

STEP 37 ? Wafer Analysis Records Removal

Date: 2026-06-29

Prompt

Remove Wafer Analysis Records table from all defect class tabs. Remove analysisRows from defect bundle and buildAnalysisRows from waferData.

Files: src/components/wafer/tabs/DefectClassTab.tsx; src/lib/waferData.ts; src/types/wafer.ts

Components: DefectClassTab (simplified)

STEP 38 ? Platform Tab Performance Optimization

Date: 2026-06-29

Prompt

Improve tab and section response time without UI visual changes. TabPanelHost lazy mount keep-alive hidden inactive tabs. Remove artificial setTimeout delays in usePlatformData. Executive dashboard useMemo not skeleton gate. React Query cache defaults refetchOnWindowFocus false staleTime 60s. Wafer heatmap pan functional setState. Apply TabPanelHost to scan-chain mbist lbist wafer-analysis cost-intelligence alerts recommendation-analysis.

Files: src/components/platform/TabPanelHost.tsx; src/hooks/usePlatformData.ts; src/app/dashboard/page.tsx; src/components/providers/QueryProvider.tsx; src/app/dashboard/scan-chain/page.tsx; src/app/dashboard/mbist/page.tsx; src/app/dashboard/lbist/page.tsx; src/app/dashboard/wafer-analysis/page.tsx; src/app/dashboard/cost-intelligence/page.tsx; src/app/dashboard/alerts/page.tsx; src/components/recommendation/RecommendationCenterContent.tsx; src/components/charts/WaferHeatmap.tsx

Components: TabPanelHost useFilteredExecutiveData QueryProvider

STEP 39 ? Record All Prompts (Full Platform Update)

Date: 2026-06-29

Prompt

Update all prompts to prompts.csv and README.md including Wafer Analysis module all tabs data images UI refinements performance optimization and current platform state.

Files: prompts.csv; README.md

Components: Prompt documentation archive (STEP 1-39)

BACKEND ? VERILUMEN Backend Implementation

Date: 2026-07-06

Prompt

VERILUMEN Backend Implementation Prompt FastAPI SQLAlchemy Postgres Redis MinIO ARQ. Phases 1-9 infra MinIO Redis auth uploads dashboard notifications. Separate verilumen-api repo. See README VERILUMEN Backend Implementation.

Files: (separate verilumen-api repo)

Components: FastAPI MinIO Redis ARQ routers workers

DATABASE ? VERILUMEN Database Schema

Date: 2026-07-06

Prompt

VERILUMEN Database Schema Alembic Phase 2 reconcile 39-table Postgres. Dims auth upload pipeline analytics recommendation_feedback indexes. See README VERILUMEN Database Schema.

Files: (separate verilumen-api repo) alembic/versions/*.py

Components: Alembic migrations SQLAlchemy models

STEP 40 ? Frontend API Client Foundation

Date: 2026-07-06

Prompt

STEP 40 Environment and typed API client. env.local.example NEXT_PUBLIC_API_URL NEXT_PUBLIC_API_MODE mock. src/lib/api client config auth dashboard uploads notifications. Thin fetch wrapper auth injection typed errors maps backend /api/v1. Default mock nothing breaks.

Files: env.local.example; src/lib/api/*.ts

Components: api client authApi dashboardApi uploadsApi notificationsApi

STEP 41 ? Frontend Auth

Date: 2026-07-06

Prompt

STEP 41 Auth. Extend userStore session tokens setSession clearSession. login page AuthGuard on dashboard layout. Replace hardcoded Alex Johnson uploadedBy with getSessionDisplayName. Live mode redirects to login.

Files: src/stores/userStore.ts; src/app/login/page.tsx; src/components/auth/AuthGuard.tsx; src/components/layout/DashboardLayout.tsx; src/components/upload/UploadDataModal.tsx; src/components/upload/UploadLogFileModal.tsx

Components: LoginPage AuthGuard session tokens

STEP 42 ? Frontend Module Data Hooks

Date: 2026-07-06

Prompt

STEP 42 Consolidate data access behind hooks. useFiltered Module Data per module recommendation alerts mbist lbist cost scan-chain wafer. Repoint 50 direct mock imports. useFilterStore returns kpis rows isLoading. Order: recommendation 10 alerts 8 mbist lbist 7 cost 7 scan-chain 6 wafer 4.

Files: (planned) src/hooks/use*ModuleData.ts

Components: useFilteredModuleData hooks

STEP 43 ? Frontend Live Dashboard Wiring

Date: 2026-07-06

Prompt

STEP 43 Live-wire dashboards. NEXT_PUBLIC_API_MODE live uses useQuery queryKey module tab filters queryFn dashboardApi. Flip flag per module as backend lands. Mock keeps useMemo path.

Files: (planned) src/hooks/usePlatformData.ts

Components: useQuery live dashboard

STEP 44 ? Frontend Filters and Search

Date: 2026-07-06

Prompt

STEP 44 Wire filters and search. Side effect of STEP 42 hooks reading useFilterStore. searchPlatform points at GET /search?q= when live keeping useGlobalSearch interface.

Files: (planned) src/lib/searchIndex.ts

Components: GlobalSearch live search

STEP 45 ? Frontend Real Upload Flow

Date: 2026-07-06

Prompt

STEP 45 Real upload flow. Replace simulateUpload with presign PUT MinIO complete SSE. Drop fileCache from uploadStore. dataHistory logHistory from GET /uploads/data /uploads/log useQuery.

Files: (planned) src/components/upload/*.tsx; src/stores/uploadStore.ts

Components: presigned upload SSE

STEP 46 ? Frontend Notifications API

Date: 2026-07-06

Prompt

STEP 46 Notifications. notificationStore seed replaced with useQuery GET /notifications useMutation markRead markAllRead Redis cached unread.

Files: (planned) src/stores/notificationStore.ts; src/components/platform/NotificationCenter.tsx

Components: notifications useQuery

STEP 47 ? Frontend Recommendation RL Feedback

Date: 2026-07-06

Prompt

STEP 47 Recommendation feedback RL signal. pattern-agent scan-debug-agent test-optimization-agent apply reject calls POST /recommendations/feedback tied to recommendation_feedback table.

Files: (planned) src/components/platform/RecommendationActionButtons.tsx

Components: recommendation feedback API

STEP 48 ? Frontend Export Report API

Date: 2026-07-06

Prompt

STEP 48 Export Report. Wire ExportMenu to GET /export/:format presigned MinIO URL browser redirect.

****Files:**** (planned) src/components/platform/ExportMenu.tsx

****Components:**** export presign URL

STEP 49 ? Scan Chain Failure Analysis KPI Trim

****Date:**** 2026-07-06

Prompt

Remove Failure Correlations and Failure Reports KPI cards from Scan Chain Failure Analysis failureAnalysisKPIs.

****Files:**** src/lib/scanChainData.ts

****Components:**** FailureKPIGrid

STEP 50 ? Record Frontend Integration Prompts

****Date:**** 2026-07-06

Prompt

Update prompts.csv and README with VERILUMEN BACKEND DATABASE prompts and frontend integration STEP 40-48 full documentation.

****Files:**** prompts.csv; README.md

****Components:**** Prompt archive STEP 40-50 BACKEND DATABASE

Part B ? Session Prompts (Build + KPI)

Session 1 ? Unified Build Sequence STEP 40?53

****Date:**** 2026-07-03

Prompt

VERILUMEN Unified Build Sequence ? merge backend + frontend prompts with dependency order. Add search, AI Diagnose, AI Optimize, recommendation feedback endpoints. Continue STEP numbering from STEP 39.

Session 2 ? Backend & Database Implementation

****Date:**** 2026-07-06

Prompt

VERILUMEN Backend & Database Implementation Prompt ? FastAPI + SQLAlchemy + Postgres/Redis/MinIO. Phases for infra, schema, auth, uploads, dashboard APIs, workers, notifications.

Session 3 ? Frontend Integration STEP 40+

****Date:**** 2026-07-06

Prompt

VERILUMEN Dashboard Frontend Integration Prompt ? API client, auth, module hooks, live dashboard wiring, filters, uploads, notifications, recommendation feedback, export.

Session 4 ? 20 Prompts to Finish Build

****Date:**** 2026-07-06

Prompt

VERILUMEN 20 Prompts to Finish the Build ? gap-analysis driven prompts for parser, live analytics, alert UI, RL consumer, STIL/WGL/PAT parsers, cost engine, deep analytics.

Session 5 ? P1-7 Enterprise File Parser

Date: 2026-07-06

Prompt

P1-7 Enterprise File Parser ? real STDF + LOG parsing in parse_worker.py. Do not rewrite project. Extend existing workers and models.

Session 6 ? Stage 7 Live Analytics

Date: 2026-07-06

Prompt

Stage 7 Live Analytics & Visualization Engine ? replace mock chart generators with PostgreSQL-driven analytics across all modules.

Session 7 ? Scan Debug KPI List

Date: 2026-07-07

Prompt

List KPI cards for Recommendation Analysis ? Scan Debug Recommendation Agent: Broken Chains Detected, Constraint Violations, Timing Debug, Power Debug, Defect Suspects, Investigation Recommendations, etc. (15 KPIs total).

Session 8 ? Scan Chain Overview Redesign

Date: 2026-07-07

Prompt

Scan Chain Overview tab should be executive summary with drill-down links ? not duplicate Pattern/Failure/Diagnosis sub-tabs. KPIs + health summary + trend analytics + mini KPI drill-down sections.

Session 9 ? Remove AI Detection Accuracy KPI

Date: 2026-07-07

Prompt

Remove AI Detection Accuracy KPI card from Scan Chain Failure Analysis.

Session 10 ? KPI Same Size Audit

Date: 2026-07-07

Prompt

Check all KPI cards are same size across complete application. Standardize all KPI cards to enterprise design system.

Session 11 ? Enterprise KPI Card Refactor

Date: 2026-07-07

Prompt

Refactor entire KPI card system into single reusable EnterpriseKPICard component.
Apply across: Scan Chain, Pattern Recommendation, Scan Debug, Test Optimization, MBIST, LBIST, Wafer, Cost Intelligence, Dashboard, Alerts.
Card: 220px height, 100% width, 22px padding, 18px radius, #111827 bg, rgba(124,58,237,.25) border.
Grid: xl 4 cols overview, xl 3 cols section, md 2 cols, sm 1 col, 24px gap.
Identical typography, icon 48x48, badge, trend, sparkline 44px bottom. Truncate overflow. w-full h-full.

Session 12 ? KPI Typography Standardization

Date: 2026-07-07

Prompt

Standardize all KPI cards typography:

Title: 16px Medium #94A3B8 | Value: 44px Bold #FFFFFF line-height 48px | Subtitle: 14px Regular #64748B

Trend: 15px SemiBold green #10B981 / red #EF4444 | Badge: 12px SemiBold height 26px padding 6px 12px rounded-full

Layout: Icon 48x48 top-left, badge top-right, title, value, subtitle or trend, sparkline 44px bottom.

Apply to Dashboard, Scan Chain, MBIST, LBIST, Wafer, Cost Intelligence, Recommendation, Alerts, Settings.

Session 13 ? KPI Text Consistency Fix

Date: 2026-07-07

Prompt

Check complete application ? KPI card text was not same in all KPI cards. Fix inconsistent value sizes and meta line styling.

Session 14 ? Fill Gaps and Save

Date: 2026-07-07

Prompt

Check all prompts, fill gaps, save what was made ? sync bd-1, commit KPI standardization work.

Part C ? Final KPI Design System (Implemented)

| Item | Specification |

|---|---|

Git commits: c1-com `bdc03e3` · bd-1/dashboard `56c8fff`